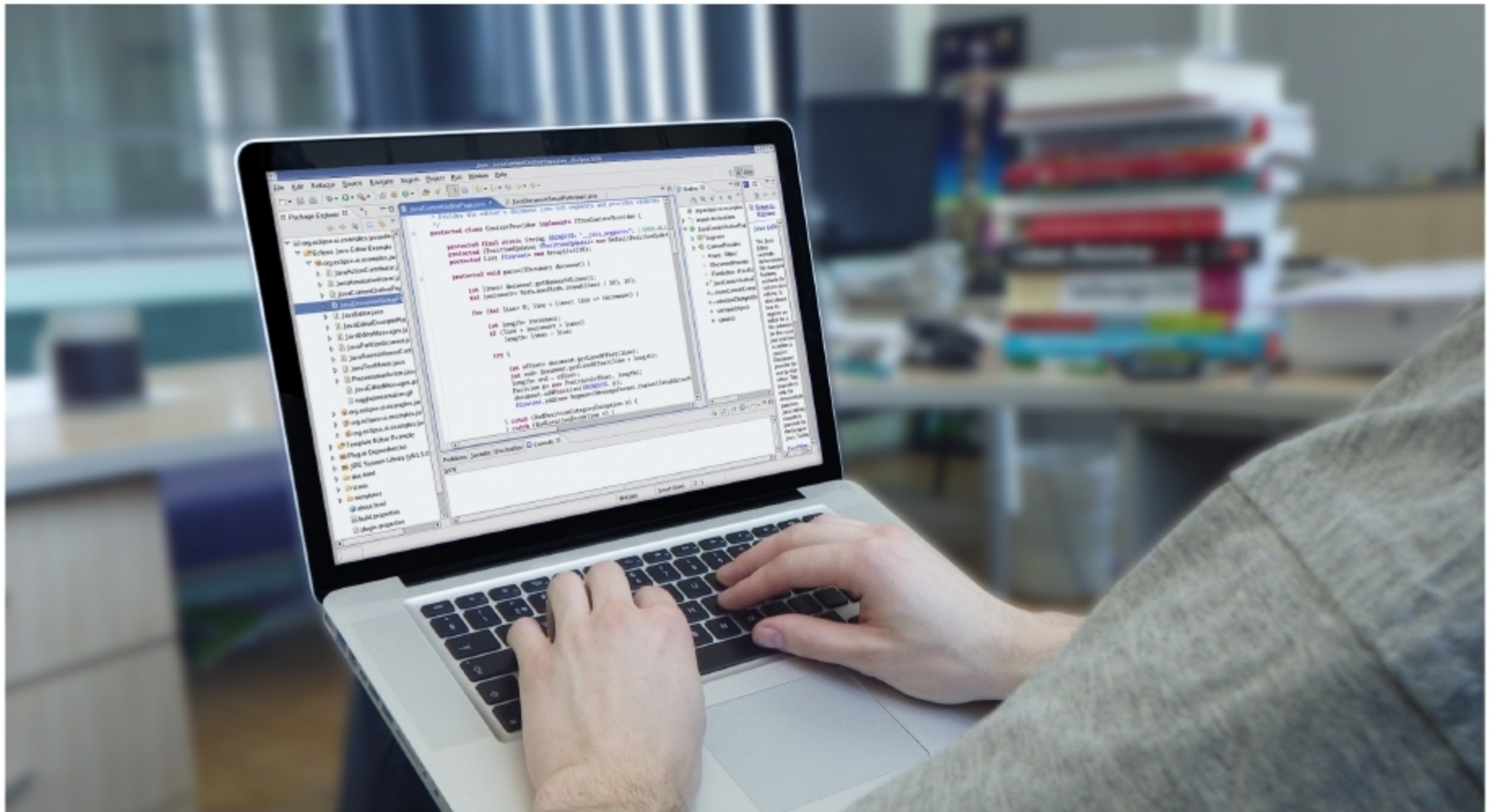


Using Eclipse with Java to Develop Software Applications

Eclipse is an integrated development environment (IDE) used when building software applications. It is one of the most popular, feature-rich IDEs in the Java programmer community, enabling rapid development and improving code quality. Eclipse supports all major programming languages like Java, C/C++ and Python. In this article, we'll use Eclipse with the Java programming language.



The good thing about Eclipse is that it's free and open source software. Developed and maintained by the Eclipse Foundation, it is released under the terms of the Eclipse Public License; so we can use it for personal as well as commercial purposes.

The salient features of Eclipse are:

1. It supports the integration of software development tools like the compiler, debugger, version control system, and so on.
2. It supports code navigation, code refactoring, auto code generation and has many more useful functions.
3. It provides a rich set of plug-ins to enhance IDE functionality further. For instance, it provides a plug-in for unit-testing frameworks.

Setting it up

Eclipse can be downloaded freely. To get the latest version, visit <https://www.eclipse.org/downloads/> and follow the on-screen instructions.

Once the downloading is done, extract the `.tar.gz` bundle using the following command:

```
$ tar xvf eclipse-inst-linux64.tar.gz
```

That's it — the installation is done! Now you're ready to use Eclipse. Just go inside the extracted directory and double-click on the executable, namely, 'eclipse'. It will launch the IDE.

Getting familiar with Eclipse

Let us get familiar with Eclipse's window. After launching Eclipse, the window

shown in Figure 1 will get opened.

In Figure 1:

1. The topmost title bar shows the workspace's name. In my case, it is 'eclipse-workspace'.
2. Below that, there is a menu bar which offers various functionalities.
3. Just below that is a toolbar that provides shortcuts like compile, run, debug and much more.
4. On the left side is a 'Package Explorer', which lists packages and classes.
5. At the bottom is the console window, where output/errors are displayed.

Don't worry if you think this is not making any sense at this moment. Later sections of this article discuss all these components, briefly.